Speech Anatomy 1

Imaging

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Seeing Speech

Maybe you think speech looks like this:

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Seeing speech

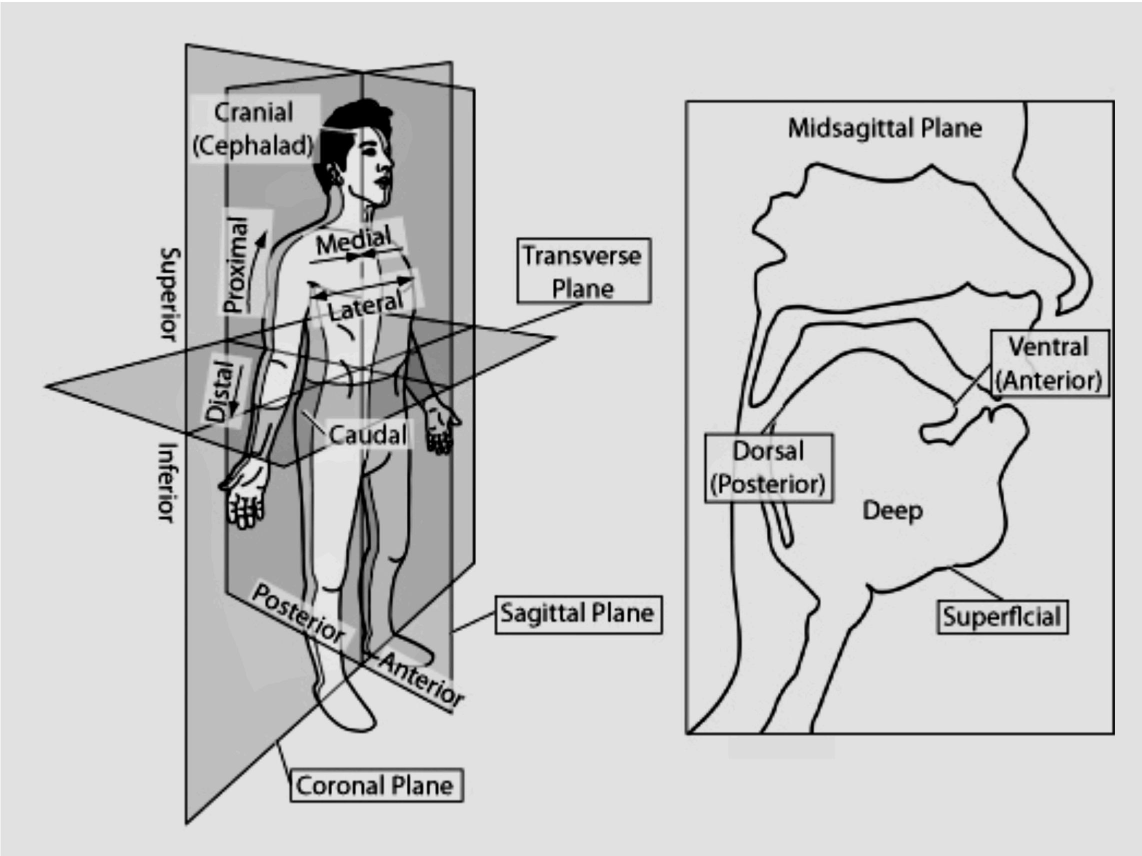
Speech is a physical activity

videos/xray.webm

Today

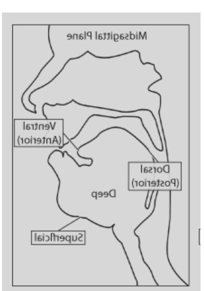
- Tools we use for looking at speech.
- Starting some anatomy.

Conceptualizing the head



Posterior	Anterior
“back”	“front”

Conceptualizing the head



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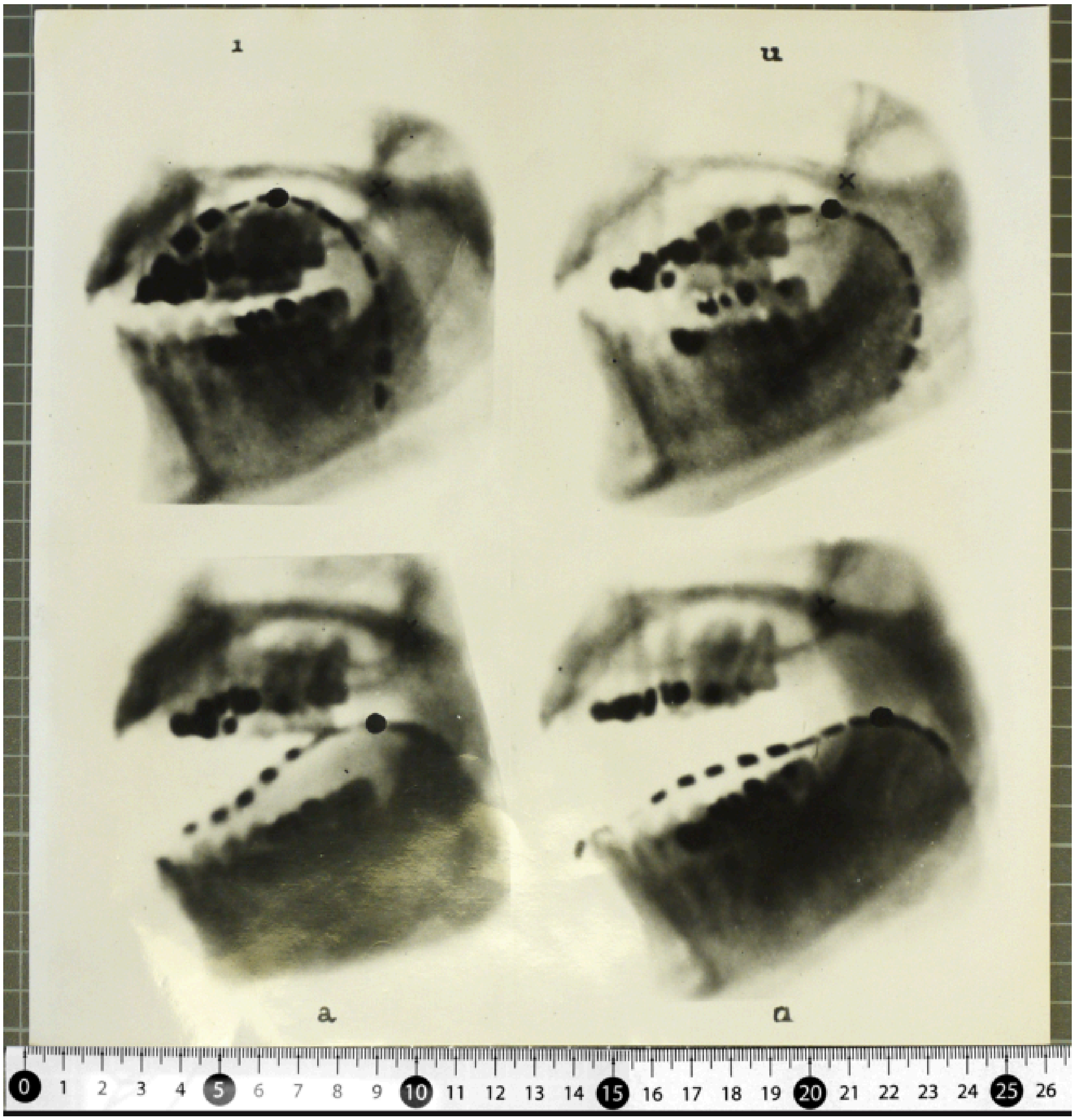
Lateral approximant				l		ɭ	ʎ	ʟ			

Tools of the trade

X-Ray

anterior ⬅️ ➡️ posterior



Tools of the trade

X-Ray

Pros?

- Better than nothing

Cons?

- Can’t image fleshy bits well
- Harmful to health
- Static/low framerate

Tools of the trade

Ultrasound tongue imaging

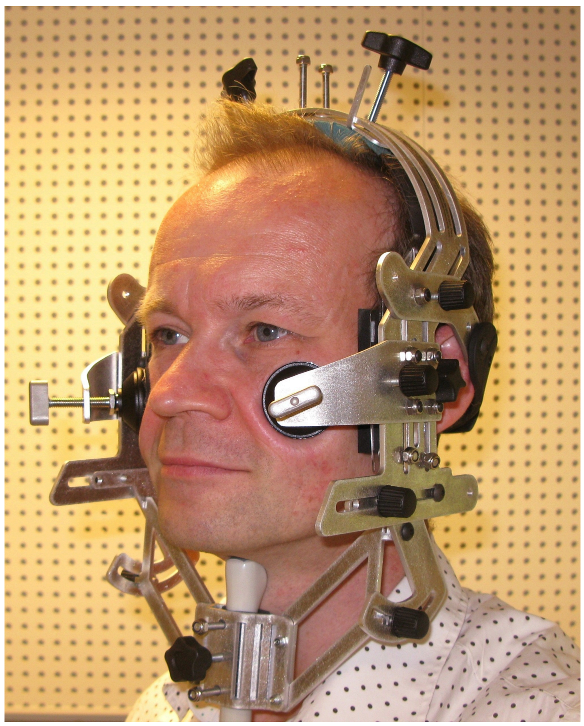


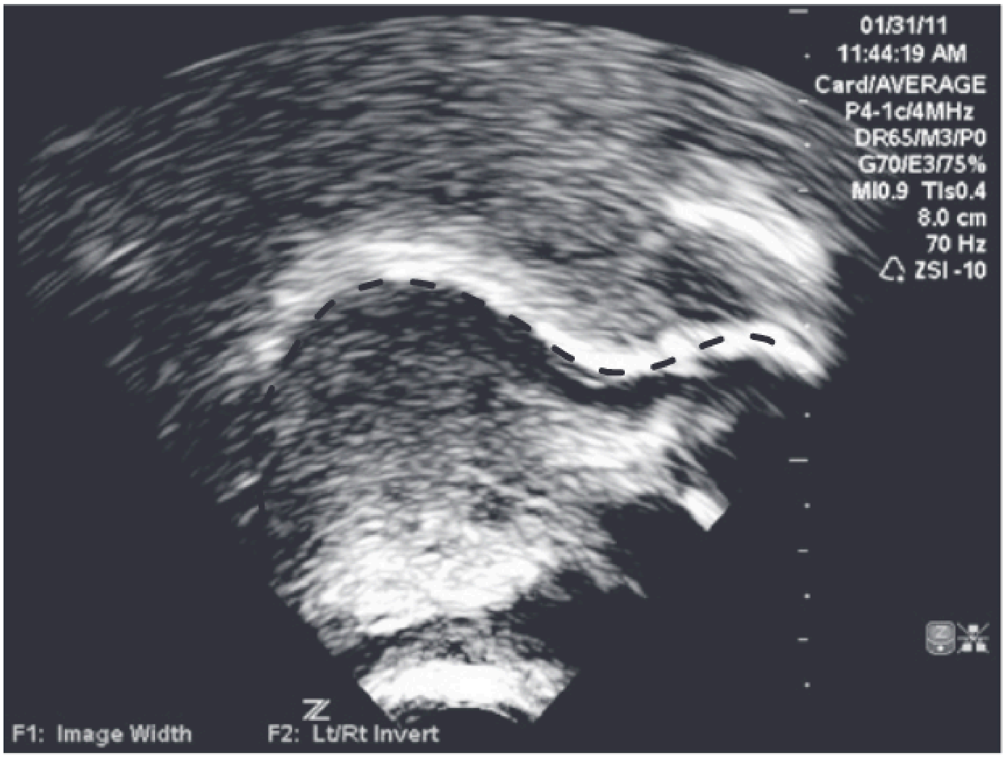
Figure 1: source: https://www.articulateinstruments.com/ultrasound-imaging/?target=Micro%2FSonoSpeech

Tools of the trade

Ultrasound tongue imaging

posterior ⬅️

➡️ anterior



S. Lin, P. S. Beddor, and A. W. Coetzee [1]

Tools of the trade

Ultrasound tongue imaging

Pros?

- Relatively non-invasive.

Cons?

- Can *only* image fleshy bits.
- Can't image across an air gap.
- Relatively low frame rate.

Tools of the trade

Electromagnetic Articulography (EMA)

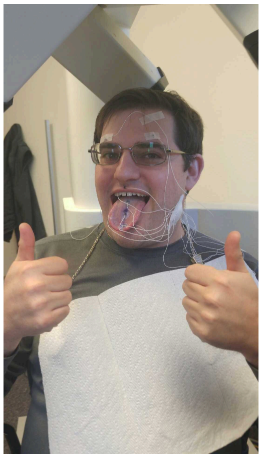
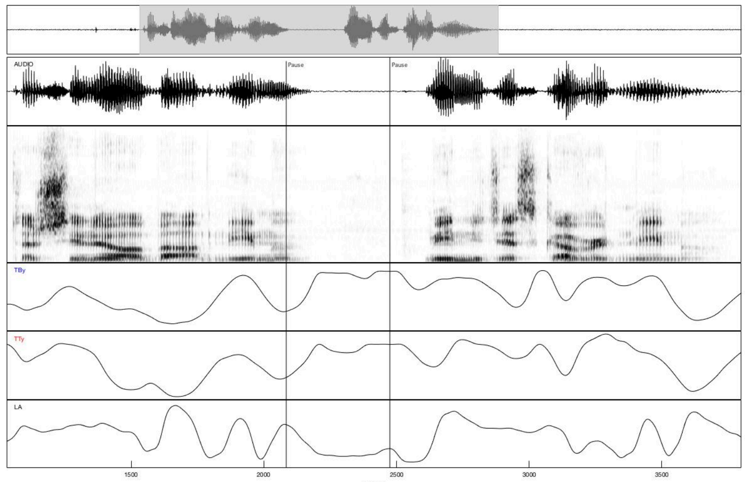


Figure 1: Credit: Will Styler



Tools of the trade

Electromagnetic Articulography (EMA)

Pros?

- Very high time resolution.
- Spatially precise.

Cons?

- Invasive.
- Harder to visualize.
- Expensive equipment.

Tools of the trade

MRI

videos/mri.mp4

posterior ⬅️

➡️ anterior

Tools of the trade

MRI

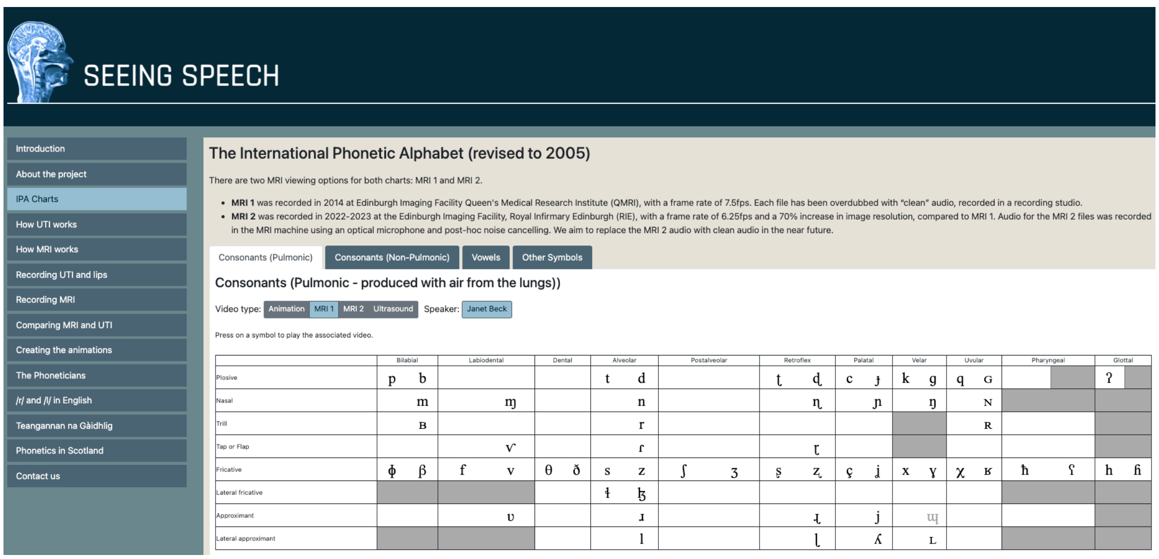
Pros?

- Non invasive.
- High quality images.

Cons?

- Low frame rate (but improving).
- *Very* loud.
- Speakers must be lying down.
- Expensive equipment.

Seeing Speech



References

Bibliography

[1] S. Lin, P. S. Beddor, and A. W. Coetzee, “Gestural reduction , lexical frequency , and sound change : A study of post-vocalic / l /,” vol. 5, no. 1, pp. 9–36, 2014, doi: 10.1515/lp-2014-0002.